Topic #:-

Diagonalization matrices

=> A square matrix A of order n over a field F is said to be diagonizable, if it is similar to a diagonal matrix over the field F.

Conditions:

1) if A has ditinct eigen values
2) if dim (V) = n and A has λI eigen vectors.
3) A is diagonizable if AM = GM

Algebraic multiplicity:

=> The order λ as a root of characteristic polynomial is called Am of λ.

Geometric multiplicity:

=> GM of an eigen value λ is nulity of $(A-\lambda I)$ or $n-\rho(A-\lambda I)$

$GM \leq AM$

Example of AM & GM:

$$A \begin{bmatrix} 1 & 0 & 2 \\ 0 & 7 & 2 \\ 0 & 0 & 9 \end{bmatrix}, \lambda=1, \lambda_2=7, \lambda=9$$

$\lambda_1 = \boxed{AM=1}$

if $\lambda_1=7$, & $\lambda_2=7$

the $\boxed{Am=2}$

G.M for $\lambda = 7$

$$|A - 7I| = \begin{bmatrix} -6 & 0 & 2 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

$n(A-7I) = 1$

$\boxed{G.M = 1}$

Definition of diagonalization.

⇒ A square matrix A is said to be diagonizable if it is similar to some diagonal matrix: that is if there exist an invertible matrix P such that

$$D = P^{-1}AP$$

is diagonal. In this case the matrix P is said to diagonalize A.

⇒ An nxn matrix A is diagonizable if A has n linearly independent eigen vectors.

$$D = \begin{bmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \ddots \\ & & & \lambda_n \end{bmatrix}$$

Diagonalization:

⇒ Suppose the n by n matrix has n linearly independent eigen vectors $x_1, \ldots, x_n$. Put them into the columns of an eigenvector matrix S. Then S^{-1} is the eigen value matrix A:

$$S^{-1}AS = \Lambda = \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix}$$

Eigenvector matrix S.

Eigenvalue matrix Λ

The matrix A is diagonalized.

Examples

Show that the following matrix A is diagonalize.

$$A = \begin{bmatrix} 4 & 2 \\ 3 & 3 \end{bmatrix}$$

Let λ be the eigen value,

so, $|A - \lambda I| = 0$

$$\left| \begin{bmatrix} 4 & 2 \\ 3 & 3 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right| = 0$$

$$\left| \begin{bmatrix} 4 & 2 \\ 3 & 3 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \right| = 0$$

$$\begin{vmatrix} 4-\lambda & 2 \\ 3 & 3-\lambda \end{vmatrix} = 0$$

$$(4-\lambda)(3-\lambda) - 6 = 0$$

$$\Rightarrow 12 - 4\lambda - 3\lambda + \lambda^2 - 6 = 0$$

$$\lambda^2 - 7\lambda + 6 = 0$$

$$\lambda^2 - 6\lambda - \lambda + 6 = 0$$

$$\lambda(\lambda - 6) - 1(\lambda - 6) = 0$$

$$(\lambda - 1)(\lambda - 6) = 0$$

$$\lambda = 1$$

$$\lambda = 6$$

Since $\{\lambda = 1, \lambda = 6\}$ are distinct so, A is diagonizable.

Eigen vectors:

for $\lambda_1 = 1$

Let $V_1 = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ be the eigen vector corresponding to $\lambda_1 = 1$.

$$(A - \lambda I) V_1 = 0$$

$$\begin{bmatrix} 3 & 2 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\sim R_2 \to R_2 - R_1 \begin{bmatrix} 3 & 2 \\ 0 & 0 \end{bmatrix}$$

$$3x_1 + 2x_2 = 0$$

$$3x_1 = -2x_2$$

Let $x_2 = 3K$

$$x_1 = \frac{-6}{3}K = -2K$$

$$V_1 = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = K' \begin{bmatrix} -2 \\ 3 \end{bmatrix} = \begin{bmatrix} 2 \\ -3 \end{bmatrix}$$

for $\lambda = 6$

Let $V_2 = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$ be the eigen vector corresponding to $\lambda_2 = 6$

$$[A - 6I]\ V_2 = 0$$

$$\begin{bmatrix} -2 & 2 \\ 3 & -3 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$-2y_1 + 2y_2 = 0$ ——— (i)

$3y_1 - 3y_2 = 0$ ——— (ii)

$$\cancel{+}2y_1 = \cancel{+}2y_2$$

$$\frac{y_1}{2} = \frac{y_2}{2}$$

$$\frac{y_1}{1} = \frac{y_2}{1} = k$$

$$V_2 = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = k \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$P = \begin{bmatrix} \cancel{-}2 & 1 \\ -3 & 1 \end{bmatrix}$$

$$P^{-1} = \frac{\text{Adj}\,P}{|P|}$$

$|P| = \cancel{-}2 + 3 = +5$

$$P^{-1}AP = \frac{1}{5}\begin{bmatrix} 1 & -1 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} 4 & 2 \\ 3 & 3 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ -3 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 6 \end{bmatrix} = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}$$

$A = \begin{bmatrix} -1 & 0 & -2 \\ 0 & 2 & 0 \\ 3 & 0 & 4 \end{bmatrix}$ is diagonalizable or not?

Eigen values:

$$\begin{vmatrix} -1-\lambda & 0 & -2 \\ 0 & 2-\lambda & 0 \\ 3 & 0 & 4-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (-1-\lambda)\begin{vmatrix} 2-\lambda & 0 \\ 0 & 4-\lambda \end{vmatrix} - 0\begin{vmatrix} 0 & 0 \\ 3 & 4-\lambda \end{vmatrix} + (-2)\begin{vmatrix} 0 & 2-\lambda \\ 3 & 0 \end{vmatrix} = 0$$

$$\Rightarrow (-1-\lambda)(2-\lambda)(4-\lambda) - 0 - 2(-(6-3\lambda)) = 0$$

$$(-\lambda-1)(2-\lambda)(4-\lambda) + 2(3)(2-\lambda) = 0$$

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$\Rightarrow (2-\lambda)[(-\lambda-1)(4-\lambda)+6] = 0$

$\Rightarrow (2-\lambda)[-4\lambda+\lambda^2-4+\lambda+6]=0$

$\Rightarrow (2-\lambda)(\lambda^2-3\lambda+2)=0$

$2-\lambda=0$	$\lambda^2-3\lambda+2=0$
$\boxed{\lambda=2}$	$\lambda^2-2\lambda-\lambda+2=0$
	$\lambda(\lambda-2)-1(\lambda-2)=0$
	$(\lambda-1)(\lambda-2)=0$
	$\boxed{\lambda=1}, \boxed{\lambda=2}$

Check Alegbraic multiplicity:

for $\lambda=1$ $AM(1)=1$, $GM=\eta(A-I)=1$

for $\lambda=2$ $AM(2)=2$, $GM=\eta(A-2I)=2$

both are same

Check GM multiplicity:

$\lambda=2$

$$(A-2I) = \begin{bmatrix} -3 & 0 & -2 \\ 0 & 0 & 0 \\ 3 & 0 & 2 \end{bmatrix}$$

$$\sim R_3 \to R_3+R_1 \begin{bmatrix} -3 & 0 & -2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Find eigen vectors:

for $\lambda=2$

$(A-2I)\,V_1=0$

$$\begin{bmatrix} -3 & 0 & -2 \\ 0 & 0 & 0 \\ 3 & 0 & 2 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\sim R_3 \to R_3+R_1 = \begin{bmatrix} -3 & 0 & -2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$x_1 = -2/3\, x_3$

$x_2 = x_2$

$x_3 = x_3$

$$\vec{V} = x_2 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + x_3 \begin{pmatrix} -2/3 \\ 0 \\ 1 \end{pmatrix}$$

are linear independent vectors of A.

for $\lambda = 1$

$(A - I I) V_2 = 0$

$$= \begin{bmatrix} -2 & 0 & -2 \\ 0 & 1 & 0 \\ 3 & 0 & 3 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Now,

$\sim R_1 \to -\frac{1}{2} R_1$

$\sim R_3 \to \frac{1}{3} R_3$
$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

$\sim R_3 \to R_3 - R_1$
$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$y_1 + y_3 = 0$

$y_1 = -y_3$

$y_2 = 0$

$y_3 = y_3$

$y_1 = -t$

$y_2 = 0$

$y_3 = t$ $\quad V_3 = t \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$

3rd L.I eigen vector of A.

$A = CDC^{-1}$

$$= \begin{bmatrix} 0 & -2/3 & -1 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & -2/3 & -1 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix}^{-1}$$

$$= \begin{bmatrix} -1 & 0 & -2 \\ 0 & 2 & 0 \\ 3 & 0 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 6 \end{bmatrix}$$

Q# Write down General procedure for diagonalizing an nxn matrix?

Step 1: Determine first whether the matrix is actually diagonalizable by searching for n linearly independent vectors.

One way to do this is to find a basis for each eigenspace and count the total number of vectors obtained if there is a total of n vectors, then the matrix is diagonizable, and if the total is less than n, then it is not.

Step:

If matrix is diagonizable then form the matrix $P = [P_1 \; P_2 \ldots P_n]$ whose column vectors are the n basis vector you obtained in step 1.

Step 3: $D = P^{-1}AP$ will be diagonizable matrix whose successive diagonal entries are the eigen values $\lambda_1, \lambda_2 \ldots \lambda_n$ that correspond to the successive columns of P.

Write down properties of eigen vectors.

Properties of eigen vector:

1) The eigen vector X of a matrix A is not unique.
2) If $\lambda_1, \lambda_2, \ldots \lambda_n$ be distinct eigen values of nxn then corresponding eigen vector $x_1, x_2, \ldots, x_n$ forms a linearly independent set.
3) If two or more eigen values are equal it may or may not be possible to get linearly independent eigen vectors corresponding to equal roots.
4) Eigen vectors of a symmetric matrix corresponding to different eigen values

are orthogonal.

Note: two eigen vectors X_1 and X_2 are called orthogonal vectors if $X_1^t \cdot X_2 = 0$ or $X_1 \cdot X_2^t = 0$

Q# Diagonalize $A = \begin{bmatrix} 1 & 1 & -2 \\ 4 & 0 & 4 \\ 1 & -1 & 4 \end{bmatrix}$ if Possible. (PP-2014)

Let $A = \begin{bmatrix} 1 & 1 & -2 \\ 4 & 0 & 4 \\ 1 & -1 & 4 \end{bmatrix}$.

$|A - \lambda I| = 0$

$$\begin{vmatrix} 1-\lambda & 1 & -2 \\ 4 & 0-\lambda & 4 \\ 1 & -1 & 4-\lambda \end{vmatrix} = 0$$

$$= (1-\lambda)\begin{vmatrix} -\lambda & 4 \\ -1 & 4-\lambda \end{vmatrix} - 1 \begin{vmatrix} 4 & 4 \\ 1 & 4-\lambda \end{vmatrix} - 2 \begin{vmatrix} 4 & -\lambda \\ 1 & -1 \end{vmatrix}$$

$= (1-\lambda)(-4\lambda + \lambda^2 + 4) - (16 - 4\lambda - 4) - 2(-4+\lambda)$

$= (1-\lambda)(\lambda^2 - 2\lambda - 2\lambda + 4) - (12 - 4\lambda) - 2(-4+\lambda)$

$= (1-\lambda)(\lambda-2)(\lambda-2) - 12 + 4\lambda + 8 - 2\lambda$

$= (1-\lambda)(\lambda-2)(\lambda-2) + 2\lambda - 4$

$= (1-\lambda)(\lambda-2)^2 + 2(\lambda-2)$

$= (\lambda-2)\{(1-\lambda)(\lambda-2) + 2\} = 0$

$= (\lambda-2)\{\lambda - 2 - \lambda^2 + 2\lambda + 2\} = 0$

$= (\lambda-2)(-\lambda^2 + 3\lambda) = 0$

$\lambda = 2$ | $-\lambda(\lambda-3) = 0$

$\lambda = 0,\ \lambda = 3$

2, 0, 3 are eigen values

Eigen vectors:

for $\lambda = 2$

$[A - 2I]V_1 = 0$

$$\begin{bmatrix} -1 & 1 & -2 \\ 4 & -2 & 4 \\ 1 & -1 & 2 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\frac{x_1}{\begin{vmatrix} -2 & 4 \\ -1 & 2 \end{vmatrix}} = \frac{-x_2}{\begin{vmatrix} 4 & 4 \\ 1 & 2 \end{vmatrix}} = \frac{x_3}{\begin{vmatrix} 4 & -2 \\ 1 & -1 \end{vmatrix}}$$

$$\frac{x_1}{0} = \frac{-x_2}{4} = \frac{x_3}{-2}$$

$$\frac{x_1}{0} = \frac{x_2}{2} = \frac{x_3}{1} = k$$

$$V_1 = k\begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix}$$

For $\lambda = 0$

$|A - 0I| V_2 = 0$

$$\begin{bmatrix} 1 & 1 & -2 \\ 4 & 0 & 4 \\ 1 & -1 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\frac{x_1}{\begin{vmatrix} 0 & 4 \\ -1 & 4 \end{vmatrix}} = \frac{-x_2}{\begin{vmatrix} 4 & 4 \\ 1 & 4 \end{vmatrix}} = \frac{x_3}{\begin{vmatrix} 4 & 0 \\ 1 & -1 \end{vmatrix}} = K$$

$$\frac{x_1}{4} = \frac{-x_2}{12} = \frac{x_3}{-4} = K$$

$$\frac{x_1}{-1} = \frac{x_2}{3} = \frac{x_3}{1} = K$$

$$V_2 = K \begin{bmatrix} -1 \\ 3 \\ 1 \end{bmatrix}$$

For $\lambda = 3$

$|A - 3I| = 0$

$$\begin{bmatrix} -2 & 1 & -2 \\ 4 & -3 & 4 \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\frac{x_1}{\begin{vmatrix} -3 & 4 \\ -1 & 1 \end{vmatrix}} = \frac{-x_2}{\begin{vmatrix} 4 & 4 \\ 1 & 1 \end{vmatrix}} = \frac{x_3}{\begin{vmatrix} 4 & -3 \\ 1 & -1 \end{vmatrix}}$$

$$\frac{x_1}{1} = \frac{-x_2}{0} = \frac{x_3}{-1} = K$$

$$V_3 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

$$P = \begin{bmatrix} -1 & 1 & 0 \\ 3 & 0 & 2 \\ 1 & -1 & 1 \end{bmatrix}$$

$$|P| = -1 \begin{vmatrix} 0 & 2 \\ -1 & 1 \end{vmatrix} - 1 \begin{vmatrix} 3 & 2 \\ 1 & 1 \end{vmatrix} + 0 \begin{vmatrix} 3 & 0 \\ 1 & -1 \end{vmatrix}$$

$= (-1)(2) - 1(3-2) + 0$

$= -2 - 1$

$A_{11} = 2$, $A_{21} = -1$, $A_{31} = 2$

$A_{12} = 1$, $A_{22} = -1$, $A_{32} = 2$

$A_{13} = -3$, $A_{23} = 0$, $A_{33} = -3$

$$\text{Adj } P = \begin{bmatrix} 2 & 1 & -3 \\ -1 & -1 & 0 \\ 2 & 2 & -3 \end{bmatrix}^t$$

$$\begin{bmatrix} 2 & -1 & 2 \\ 1 & -1 & 2 \\ -3 & 0 & -3 \end{bmatrix}$$

$$= \frac{-1}{3} \begin{bmatrix} 2 & -1 & -2 \\ 1 & -1 & 2 \\ -3 & 0 & -3 \end{bmatrix} \begin{bmatrix} 1 & 1 & -2 \\ 4 & 0 & 4 \\ 1 & -1 & 4 \end{bmatrix} \begin{bmatrix} -1 & 1 & 0 \\ 3 & 0 & 2 \\ 1 & -1 & 1 \end{bmatrix}$$

$$= \frac{-1}{3} \begin{bmatrix} 2 & -1 & 2 \\ 1 & -1 & 2 \\ -3 & 0 & -3 \end{bmatrix} \begin{bmatrix} 0 & 3 & 0 \\ 0 & 0 & 4 \\ 0 & -3 & 3 \end{bmatrix}$$

$$\frac{-1}{3} \begin{bmatrix} 0 & 0 & 0 \\ 0 & -6 & 0 \\ 0 & 0 & -9 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

D

Q# Find a matrix P that diagonalize A, where $A = \begin{bmatrix} 2 & 0 & -2 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$ (PP-2020)

Let $A = \begin{bmatrix} 2 & 0 & -2 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$

$|A - \lambda I| = 0$

$\because$ it is triangular matrix so the diagonal values are eigen values

So, 2, 3, 3 are eigen values.

Eigen vectors:

for $\lambda = 2$

$|A - 2I| \, V_1 = 0$

$$\begin{bmatrix} 0 & 0 & -2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\frac{x_1}{\begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix}} = \frac{-x_2}{\begin{vmatrix} 0 & 0 \\ 0 & 1 \end{vmatrix}} = \frac{x_3}{\begin{vmatrix} 0 & 1 \\ 0 & 0 \end{vmatrix}}$$

$$\frac{x_1}{1} = \frac{-x_2}{0} = \frac{x_3}{0} = k$$

let ~~$x_1 x_2$~~ 1:

$V_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

for $\lambda = 3$

$|A - 3I| V_2 = 0$

$$\begin{bmatrix} -1 & 0 & -2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$\cdot\ -x_1 - 2x_3 = 0 \Rightarrow x_1 = -2x_3$

let $x_2 = x_2$

$x_3 = x_3$

$$V_2 = \begin{bmatrix} -2x_3 \\ x_2 \\ x_3 \end{bmatrix} = x_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix}$$

$$P = \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$|P| = 1 \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} - 0 \begin{vmatrix} 0 & -2 \\ 0 & 1 \end{vmatrix} - 0 \begin{vmatrix} 0 & -2 \\ 1 & 0 \end{vmatrix}$$

$= 1(1) - 0 + 0 = 1$

$A_{11} = 1$, $A_{21} = 0$, $A_{31} = 2$

$A_{12} = 0$, $A_{22} = 1$, $A_{32} = 0$

$A_{13} = 0$, $A_{23} = 0$, $A_{33} = 1$

$$\text{Adj}\,P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix}^t = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$D = P^{-1}AP$

$$= (1) \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 & -2 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 & -6 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

Q# Show that the matrix

$$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$$ cannot be diagonalized.

∴ it is triangular matrix the eigen values will be the main diagonal values so,

eigen values are $0, 1, 1$

Eigen vector:

for $\lambda = 0$

$|A - 0I| v_1 = 0$

$$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Now,

$$R_3 \to R_3 - R_1 \quad \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$x_3 = 0$

$x_2 + 2x_3 = 0$

$x_2 = -2x_3 \Rightarrow \boxed{x_2 = 0}$

Let $x_1 = t$

$$v_1 = \begin{bmatrix} t \\ 0 \\ 0 \end{bmatrix} = t \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

for $\lambda = 1$

$|A - I| v_2 = 0$

$$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Now,

$$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

$$R_2 \to R_2 - 2R_1 \quad \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$x_3 = 0$

Let $x_2 = x_2$

$x_1 = x_1$

$$V_2 = \begin{bmatrix} 0 \\ x_2 \\ x_1 \end{bmatrix} = x_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + x_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$P = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$|P| = 1 \begin{vmatrix} 1 & 0 \\ 0 & 0 \end{vmatrix} - 0 \begin{vmatrix} 0 & 1 \\ 0 & 0 \end{vmatrix} - 0 \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix}$$

$$= 0 - 0 - 0$$

$$= 0 \neq \text{singular}$$

So, we cannot find P^{-1} so, the matrix A is not diagonizable.